

SESSION 2

RAD-447 Radionuclide Imaging

Gamma Camera, Planar Imaging & SPECT

From single gamma photons to two-dimensional and tomographic images

Course	RAD-447 Radionuclide Imaging / Nuclear Medicine
Date	2026
Lecturer	A. A. Zubair
Suggested study time	180–240 minutes

Learning Objectives

By the end of this session, the student should be able to:

1. identify the major components of a conventional gamma camera;
2. explain how a collimator provides directional information about incoming gamma photons;
3. describe the roles of the NaI(Tl) scintillation crystal, photomultiplier tubes and positioning electronics;
4. explain why pulse-height analysis and an energy window are used;
5. distinguish planar scintigraphy from SPECT; and
6. compare filtered back projection with iterative SPECT reconstruction at a conceptual level.

Key Concept

A gamma camera does not simply “photograph” radioactivity. Gamma photons leave the patient in many directions and contain no visible label showing where they originated. The camera must therefore select useful photon directions, convert deposited gamma energy into light, convert that light into electrical signals, estimate event position and energy, and finally accumulate many accepted events into an image.

1 Session Orientation

Session 1 established that radionuclide imaging depends on radioactive decay occurring inside the patient. Session 2 now follows the emitted gamma photon from the patient to the final image. The central question is: **how can a detector determine both the position and the energy of a photon event when gamma photons arrive from many directions?**

A useful way to organize the entire gamma-camera process is:

direction selection → energy conversion → signal amplification → position/energy calculation → event acceptance → image formation

Why this sequence matters

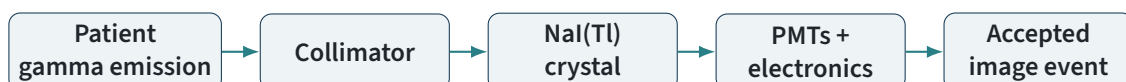
Each component solves a different imaging problem. The collimator supplies directional information, the crystal converts gamma energy into light, the PMTs/electronics estimate position and energy, and the computer accumulates accepted events. If one stage performs poorly, image quality can deteriorate even when the radiopharmaceutical distribution in the patient is unchanged.

2 The Gamma Camera: From Photon to Image

The **gamma camera** is the principal detector system used for conventional planar nuclear medicine and for SPECT. It is designed to detect individual gamma photons emitted from a radiopharmaceutical within the patient and to determine where accepted events should be placed in the image.

The conventional gamma camera is built around six major functional components:

1. collimator;
2. scintillation crystal;
3. photomultiplier tubes (PMTs);
4. positioning electronics;
5. pulse-height analyzer; and
6. computer / image-processing system.



Each accepted event contributes a count to a particular image location. A diagnostic image is therefore built from a very large number of individual detected events rather than from a single exposure in the same sense as conventional radiography.

This count-based nature is fundamental. A region that receives more accepted events is represented by a larger number of counts. Because radioactive decay and photon detection are statistical processes, images with fewer counts appear noisier, whereas images with more counts generally have better count statistics. The practical balance between counts, acquisition time and administered activity is developed further in the image-quality session.

Clinical / Practical Link

A gamma camera must solve two questions for every useful photon event: **Where did the photon interact?** and **Was its measured energy consistent with the radionuclide being imaged?** Positioning electronics answer the first question; energy discrimination helps answer the second.

3 The Collimator: Giving Direction to Gamma Photons

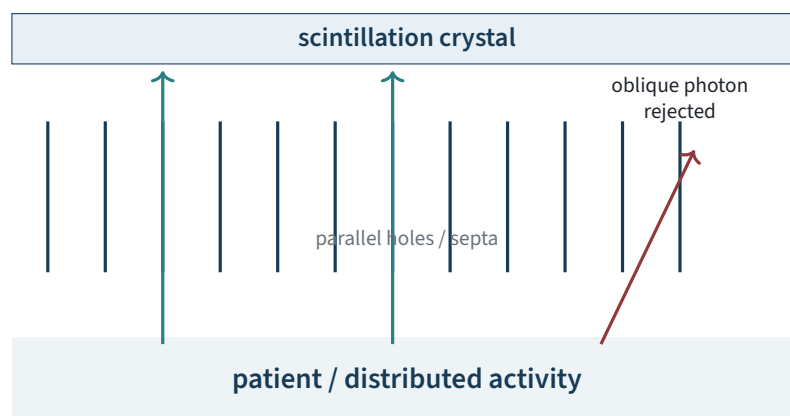
The **collimator** is positioned in front of the scintillation crystal. Its fundamental function is to **select photons according to their direction of travel**. This is necessary because gamma photons emitted within the patient travel in many directions.

Without directional selection, a photon detected at one point on the crystal could have originated from many possible positions inside the patient. The resulting image would contain very poor spatial information.

3.1 Parallel-hole collimation

The parallel-hole collimator is the most commonly used general-purpose design in conventional gamma-camera imaging. It contains many approximately parallel channels separated by dense septa. Photons traveling along suitable directions can pass through the holes and reach the crystal, whereas many obliquely traveling photons are absorbed by the collimator.

The walls between adjacent holes are called **septa**. Their job is to absorb photons traveling at unsuitable angles. Collimator geometry therefore acts as a form of **mechanical spatial encoding**: the camera accepts only a restricted set of photon paths so that the point at which a photon reaches the detector is more closely related to where it originated in the patient.



Other important collimator designs include:

Collimator	General idea
Parallel-hole	Many parallel channels; commonly used for general imaging and approximately maintains image size.
Pinhole	Uses a small aperture and is useful when magnification of a small structure is desirable.
Converging	Channels angle inward so that a selected field can be magnified at the detector.
Diverging	Channels angle outward, allowing a larger field to be represented on a detector surface.

Exam focus

The collimator is a **direction selector**, not the device that converts gamma radiation into light. The scintillation crystal performs the gamma-to-light conversion.

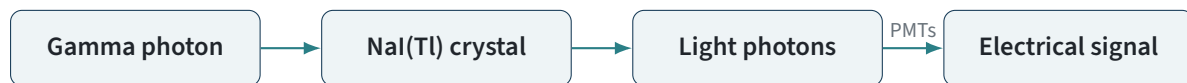
Resolution-sensitivity trade-off

A collimator that accepts a narrower range of photon directions can improve spatial localization, but fewer photons reach the crystal. A design that accepts more photons improves sensitivity but usually sacrifices some spatial precision. This trade-off between **resolution** and **sensitivity** is a recurring principle in nuclear medicine detector design.

In practice, this means there is no single “best” collimator for every examination. Hole length, hole diameter, septal thickness and the photon energy of the radionuclide all influence performance. The selected collimator must therefore match the clinical task and the radionuclide being imaged.

4 The Scintillation Crystal

The conventional gamma camera commonly uses **sodium iodide activated with thallium**, written as **NaI(Tl)**. When a gamma photon interacts within the crystal, some of its energy is converted into a small burst of visible or near-visible light.



The amount of light produced is related to the energy deposited in the crystal. This matters because the camera subsequently uses the measured pulse size as an estimate of photon energy.

NaI(Tl) is efficient for many conventional gamma-camera photon energies, but it is also hygroscopic, meaning that it can absorb moisture and must be sealed from the environment. The crystal is made sufficiently thick to absorb useful photons while preserving acceptable spatial resolution.

Common Error

Do not describe the scintillation crystal as “detecting the location by itself.” It converts deposited gamma energy into light. Event position is estimated from the pattern and relative sizes of signals measured by multiple photomultiplier tubes and the associated electronics.

5 Photomultiplier Tubes and Positioning Electronics

A scintillation flash is weak. **Photomultiplier tubes** convert the light into electrical signals and amplify them. Because several PMTs receive different fractions of the light produced by one interaction, the electronics can use their relative outputs to estimate the event’s position on the crystal.

The system derives three clinically important quantities:

- an **X coordinate**;
- a **Y coordinate**; and
- an estimate of the event’s **energy**.

Conceptually, stronger signals from PMTs nearest the interaction pull the calculated position toward that region. The total signal is used as an estimate of how much energy was deposited.

This weighted-position calculation is often referred to conceptually as **Anger logic**. Students do not need the circuit mathematics here; the key idea is that **relative PMT signal strengths provide position**, while the **combined signal provides an energy estimate**.

Position and energy are calculated from the same event

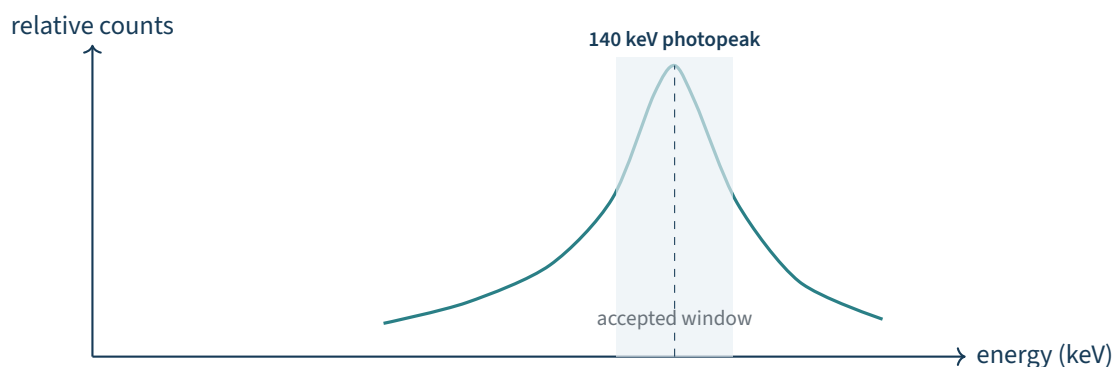
The camera must know both **where** to place the event in the image and whether its **energy** is suitable for acceptance. This is why PMTs, positioning circuits and energy discrimination work as one chain rather than as isolated components.

6 Pulse-Height Analysis and the Energy Window

For Tc-99m imaging, the principal photopeak is approximately **140 keV**. The camera measures the pulse produced by each event and compares it with an accepted energy range, called the **energy window**.

Events outside the chosen window may be rejected. This helps reduce contributions from photons that have lost energy through scattering or from other unwanted radiation.

For Tc-99m, a common teaching example is a window centered on the 140-keV photopeak. A nominal 20% window, for example, accepts energies over a band around the photopeak rather than one exact value. The precise window used depends on protocol, equipment and quality-control requirements.

**Clinical / Practical Link**

Scattered photons may still reach the detector, but their directions no longer reliably indicate the original emission path and their measured energies are often lower. Energy discrimination therefore improves image quality by rejecting many events that would otherwise add misleading counts.

The most important mechanism is **Compton scatter**: the photon changes direction and loses part of its energy during interaction with tissue. This is why energy information helps the camera distinguish many scattered events from photons that reach the detector without substantial energy loss.

7 Planar Scintigraphy

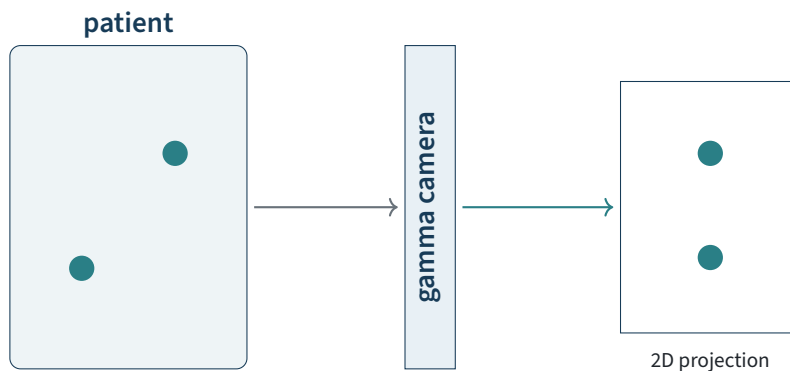
In **planar imaging**, the camera acquires a two-dimensional projection from a fixed view. The image represents the distribution of accepted counts as seen from that direction.

Common examples include:

- anterior or posterior bone images;
- thyroid imaging;
- lung perfusion imaging; and
- renal imaging.

The major limitation is **superimposition**. Activity from structures at different depths may overlap in the same two-dimensional projection, making precise localization more difficult.

Planar imaging can still be highly useful because it is simple, relatively fast and can cover a large field of view. Its limitation is primarily depth localization: a single projection compresses three-dimensional tracer distribution into two dimensions.



8 SPECT: Adding Tomographic Information

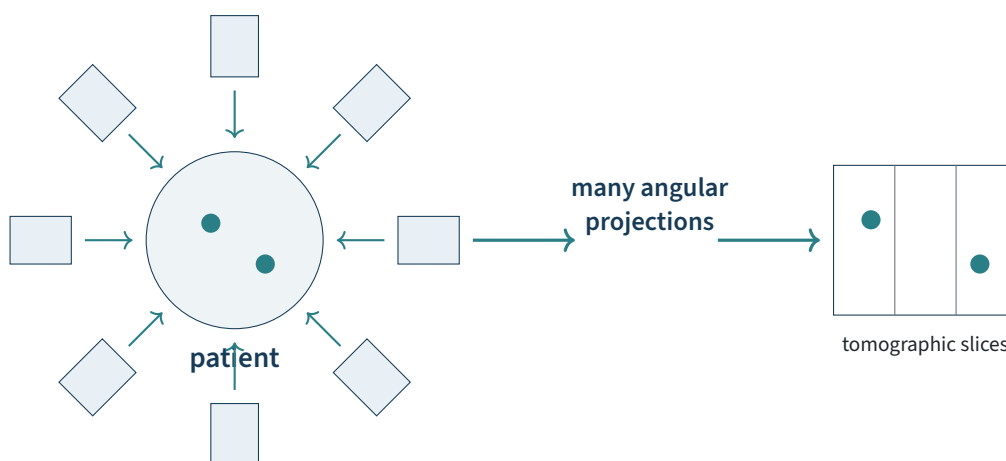
SPECT stands for **Single Photon Emission Computed Tomography**. Instead of acquiring only one fixed projection, a gamma camera rotates around the patient and acquires multiple projections from different angles.

The basic chain is:



Because the projections are acquired from many angles, the reconstruction process can estimate how activity is distributed through slices rather than simply displaying all depths on top of one another.

During acquisition, the detector head or heads step or rotate through a defined angular range and record projection images at many positions. The number of projections, angular spacing, acquisition time per projection and detector radius all influence the quality of the reconstructed study. Keeping the detector as close to the patient as practical usually improves spatial resolution.



Planar versus SPECT

Planar imaging is a two-dimensional projection and is affected by superimposition. **SPECT** acquires many projections around the patient and reconstructs them into tomographic slices and 3D datasets.

9 SPECT Image Reconstruction

Two broad reconstruction approaches are important at this level.

9.1 Filtered back projection

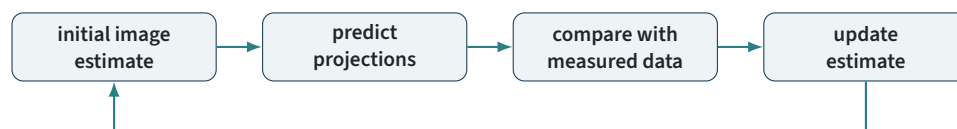
Filtered back projection (FBP) is a traditional reconstruction method. In simple terms, projection information is mathematically spread back through the image space along the directions from which it was acquired. Filtering is used to reduce the blurring that would occur with simple back projection alone.

FBP is computationally direct and historically important, but it assumes a relatively simple imaging model. It is sensitive to noisy or incomplete projection data and does not naturally model every physical effect that degrades SPECT images.

9.2 Iterative reconstruction

Two commonly encountered iterative methods are **MLEM** and **OSEM**. Instead of producing the final image in one direct mathematical step, an iterative method starts with an estimate, predicts what projections that estimate would produce, compares them with the measured data, and repeatedly updates the estimate.

Because this process explicitly compares predicted and measured data, additional information about attenuation, collimator-detector response, scatter or noise behaviour can be incorporated into the reconstruction model when the system and software support it.



Iterative methods can incorporate more realistic models of the imaging system and can handle attenuation, scatter and noise more flexibly than traditional FBP.

MLEM and OSEM conceptually

MLEM updates an image estimate using the complete measured dataset during each iteration. **OSEM** accelerates the general idea by dividing the projection data into subsets and updating the estimate more frequently. Students do not need the full mathematics at this stage; the important concept is that these methods repeatedly refine an estimated activity distribution so that it better explains the measured projection data.

10 Planar and SPECT Acquisition in Practice

A good nuclear-medicine image depends on both detector physics and acquisition technique. Before imaging, the operator confirms the correct radionuclide, collimator and energy window, positions the patient and detector appropriately, and selects acquisition parameters such as matrix size, count target or acquisition time. In SPECT, additional choices include angular range, number of projections, detector orbit and time per projection.

Radiographer / technologist perspective

Poor positioning, excessive detector-to-patient distance, patient movement, an incorrect collimator or an incorrectly centred energy window can degrade the study even when the tracer preparation and administration were correct. Equipment settings and patient handling are therefore part of image formation, not separate from it.

11 Why Acquisition Geometry Matters

SPECT reconstruction depends on having projections that represent the patient from many directions. If a projection is missing, severely degraded by motion, or inconsistent with the others, the reconstruction may be affected.

This makes patient cooperation and stable positioning particularly important during tomographic acquisition. The later image-quality and quality-control session will examine motion, attenuation, scatter, equipment problems and other artifacts in more detail.

Common Error

Do not say that SPECT “takes cross-sectional pictures directly.” The gamma camera first acquires multiple two-dimensional projections. The cross-sectional images are subsequently **reconstructed computationally** from those projections.

Key Takeaways

- The gamma camera detects single gamma photons emitted from radiopharmaceuticals in the patient.
- The collimator provides directional selection; the NaI(Tl) crystal converts deposited gamma energy into light.
- PMTs convert and amplify scintillation light, while electronics estimate X position, Y position and event energy.
- Pulse-height analysis and an energy window help reject many scattered or unwanted events.
- Planar scintigraphy produces a 2D projection and therefore suffers from superimposition.
- SPECT acquires multiple angular projections and reconstructs them into tomographic images.
- FBP is a traditional reconstruction method; MLEM and OSEM are iterative approaches.

Key Terms

Term	Meaning
Gamma camera	Detector system used for conventional planar radionuclide imaging and SPECT.
Collimator	Direction-selecting structure placed in front of the scintillation crystal.
NaI(Tl)	Thallium-activated sodium iodide scintillation crystal commonly used in conventional gamma cameras.
Photomultiplier tube	Device that converts scintillation light into an amplified electrical signal.
Energy window	Accepted range of measured photon energies used for event discrimination.
Planar scintigraphy	Fixed-view two-dimensional projection imaging.
SPECT	Single Photon Emission Computed Tomography; tomographic imaging from multiple gamma-camera projections.
FBP	Filtered back projection, a traditional tomographic reconstruction approach.
MLEM / OSEM	Iterative reconstruction methods used to refine an estimated activity distribution from projection data.

Self-Assessment Questions

1. List the six major components of a conventional gamma camera described in the lecture.
2. Explain why a collimator is required before the scintillation crystal.
3. State the role of the NaI(Tl) crystal and distinguish it from the role of a PMT.
4. What three quantities are estimated from PMT/electronic signals for an accepted event?
5. Why is an energy window used when imaging Tc-99m?
6. Compare planar scintigraphy with SPECT in terms of data acquisition and image output.
7. Describe the basic SPECT workflow from gamma emission to tomographic image.
8. Distinguish filtered back projection from iterative reconstruction conceptually.

Series Navigation

1	Foundations of Radionuclide Imaging
2	Gamma Camera, Planar Imaging & SPECT — this session
3	PET, Attenuation Correction & Hybrid Imaging
4	Clinical Radionuclide Imaging Applications
5	Image Quality, Radiation Protection, QC, Artifacts & Future Directions

